

Remaining Useful Life Prediction on N-CMAPSS Turbofan Simulations: A Deep-Learning Pipeline for Variable-Length Flights

Prahalad Talur

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Abstract

Predicting remaining useful life (RUL) from turbofan sensor trajectories is a core benchmark problem in aerospace prognostics, but simulation benchmark results should be separated from claims about certified in-service use. The N-CMAPSS dataset [1] combines a commercial engine simulation model with real recorded flight envelopes, yielding variable-length flights of 10^4 – 10^5 timesteps per engine and heterogeneous operating conditions. We present a deep-learning pipeline for variable-length N-CMAPSS flights and compare recurrent, temporal-convolutional, WaveNet, and multi-scale attention families under matched preprocessing, training, and evaluation code. In a three-seed controlled FD1 suite, WaveNet is the strongest and most stable model at $T=1000$ (RMSE 6.868 ± 0.299), MSTCN is competitive but lower (RMSE 7.890 ± 0.526), and CNN-GRU is unstable because two of three seeds collapse. A matched WaveNet sweep shows that short windows are competitive across 100–1000 timesteps, with $T=500$ best by mean RMSE (6.690 ± 0.173). In a separate targeted seed sweep, the best WaveNet run reaches RMSE 6.197, $R^2 = 0.9175$, and Accuracy@20 99.71 on FD1, slightly lower than the 6.24 N-CMAPSS validation RMSE anchor reported by Solís-Martín et al. [15]; the protocols are not directly matched. Asymmetric MSE is modestly better than symmetric MSE in this suite (RMSE 6.868 ± 0.299 versus 7.214 ± 0.559), while asymmetric Huber is high variance.

1 Introduction

Modern turbofan datasets contain dense sensor suites whose readings, integrated over many cycles, encode gradual component degradation. Translating those recordings into a calibrated estimate of remaining useful life is the central task of data-driven engine prognostics. The cost structure of this task is asymmetric: conservative early predictions incur maintenance overhead, while late predictions are more serious in a safety-critical setting. The PHM-2008 scoring function [14] encodes this asymmetry directly and remains a common evaluation metric in the field. Our experiments study this problem in simulation data only and should not be read as evidence for operational deployment.

For more than a decade, deep-learning research on this problem has been driven by the synthetic CMAPSS benchmark [14], where each engine produces a few hundred cycles of low-dimensional sensor data under a small set of operating conditions. The N-CMAPSS dataset of Arias-Chao et al. [1] was introduced to close the gap between this benchmark and the conditions encountered in commercial operation: trajectories are simulated from real recorded flight envelopes spanning ascent, cruise, and descent, with sample-rate-level sensor data on the order of 10^4 – 10^5 timesteps per cycle. The hybrid physics-augmented baseline of Arias-Chao et al. [2] and the published submissions

to the 2021 PHM Society Data Challenge [12, 6, 15] establish N-CMAPSS as the contemporary benchmark of record for turbofan RUL.

N-CMAPSS poses three difficulties that did not arise on CMAPSS. Flights are variable-length, with each engine yielding a different total cycle count and per-cycle sequence length. Operating conditions are heterogeneous within and across flights, so a single model must generalize across the full envelope. And the sequence regime is qualitatively different: at 10^4 – 10^5 timesteps per cycle, naively ingesting full flights pushes recurrent and quadratic-attention models toward their failure modes for long inputs.

The methodological response to these difficulties has spanned four families. Recurrent networks such as LSTM [8], GRU [4], and bidirectional variants were the first applied. Temporal convolutional networks [3], WaveNet-style dilated convolutions [16], and dilated/inception variants [12, 6] were introduced to handle long inputs without sequential bottlenecks. Self-attention models, building on the Transformer [17] and long-sequence variants such as Informer [19], have been adapted to RUL through hierarchical and two-stage designs [7, 10]. Most recently, multi-scale TCN architectures with attention modules over the dilation hierarchy [18] have been proposed as a way to combine the long effective receptive field of TCNs with the cross-channel and cross-scale weighting of attention. The publicly reported benchmarks across these families are not directly comparable: most are reported on CMAPSS rather than N-CMAPSS, training protocols and input lengths vary substantially, and few studies report the same accuracy, RMSE, and PHM-score triple under a single harness.

This paper contributes a deep-learning pipeline for variable-length N-CMAPSS flights together with a shared registry of 22 architectures drawn from recurrent, convolutional, hybrid, transformer, and multi-scale attention families. The pipeline standardizes sequence preparation (last- N truncation, per-feature standardization), loss choices, evaluation metrics (RMSE, R^2 , PHM score, accuracy at ± 20 cycles), and the experimental harness used for head-to-head numbers. The results support the following bounded claims:

- **C1 (sequence length).** In a matched WaveNet sweep, short windows from 100 to 1000 timesteps all work well, with $T=500$ best by three-seed mean RMSE and $T=250$ close behind. The result supports short-window modeling, not a claim that exactly 1000 timesteps is optimal.
- **C2 (architecture stability).** WaveNet is the strongest FD1 model in the three-seed controlled suite. MSTCN is stable and competitive, but not dominant. CNN-GRU has one good seed and two collapsed seeds, so its single-seed rank is not reliable.
- **C3 (loss choice).** Asymmetric MSE slightly improves the three-seed WaveNet mean over symmetric MSE and asymmetric Huber in this setup, but the loss comparison remains preliminary because it covers one architecture, one dataset subset, and only three seeds.

The rest of the paper is organized as follows. Section 2 reviews related work in N-CMAPSS prognostics, attention-based RUL models, and asymmetric loss design. Section 3 describes the pipeline, including data handling, the asymmetric loss, and the MSTCN architecture. Section 4 presents the apples-to-apples benchmark across the registered architecture set. Section 5 discusses C1–C3 and the missing ablations. Section 6 concludes.

2 Related Work

We organize prior work into four families directly relevant to the claims in this paper: the N-CMAPSS dataset and its baseline competitions, attention/transformer architectures for RUL, multi-scale temporal-convolution methods, and asymmetric losses for safety-critical prognostics. We close by situating our contribution against this literature, noting up front that nearly all of the attention- and multi-scale architectures we benchmark have, to date, been published only on the older C-MAPSS dataset and that head-to-head comparability with our N-CMAPSS results is therefore limited.

2.1 N-CMAPSS dataset and baseline competitions

The N-CMAPSS dataset [1] extends the original C-MAPSS turbofan simulator [14] to run-to-failure trajectories generated under recorded flight conditions, producing variable-length flights on the order of 10^4 – 10^5 timesteps per unit. The 2021 PHM Society Data Challenge introduced the first comparable leaderboard on this dataset. The challenge winner [12] proposed a dilated-convolutional network explicitly designed to ingest variable-length input sequences with a learned condition-normalization step, which is the closest prior art to the variable-length flight handling we adopt. The second-place entry [6] used inception modules to capture multiple temporal receptive fields from windowed N-CMAPSS data, and the third-place entry [15] reported a stacked DCNN trained on $L_w = 161$ windows, achieving validation L2 RMSE 6.24 and test NASA score 3.651. Outside the challenge, a hybrid physics-plus-deep-learning baseline [2] reports an extended prediction horizon over a purely data-driven model on a fleet of nine N-CMAPSS engines, and the ANN-Flux baseline of Cohen et al. [5] reports RMSE 7.75 / NASA score 4.34 on N-CMAPSS. These are the primary numerical anchors for “what is a good N-CMAPSS result,” and they collectively establish a precedent that compact windowed inputs are the operating regime of choice on this dataset.

2.2 Attention and transformer architectures for RUL

The transformer [17] replaces recurrence with self-attention as the primary sequence-modeling primitive, and Informer [19] adapts it to long sequences via ProbSparse attention and self-attention distillation, motivating input compression rather than raw long inputs. Two recent attention models are directly relevant. The STAR transformer [7] is a two-stage hierarchical encoder–decoder that applies sequential sensor-wise and temporal attention; it operates on 32–64-step inputs and reports RMSE 10.61 / 13.47 / 10.71 / 15.87 on C-MAPSS FD001–FD004. TTSNet [10] fuses a transformer branch, a TCN branch and a multi-head self-attention branch over a noise-smoothed input and reports RMSE 11.02 / 13.25 / 11.06 / 18.26 on the same C-MAPSS subsets. Both are evaluated only on C-MAPSS, so their RMSE numbers are not directly comparable to ours: N-CMAPSS sequences are one to two orders of magnitude longer per unit, and the resulting RUL distributions differ.

2.3 TCN and multi-scale temporal methods

The temporal convolutional network [3] establishes dilated causal convolutions as a competitive sequence-modeling alternative to recurrent networks, with longer effective memory and faster training. The closest prior work to the MSTCN architecture evaluated here is the attention-based multi-scale TCN of Xu, Zhang and Miao [18], which combines a multi-scale TCN backbone with self-attention at the head and a global-fusion-attention block at the tail. Their architecture is essentially the same family represented by our MSTCN implementation and is therefore the strongest

direct precedent for our multi-scale temporal-attention comparison. As with the transformer literature, however, [18] reports results on C-MAPSS rather than N-CMAPSS, so the architectural lineage is direct but the numerical comparison with our results is not.

2.4 Loss functions for safety-critical RUL

The dominant evaluation metric for RUL on C-MAPSS and N-CMAPSS is the asymmetric exponential PHM score introduced by Saxena et al. [14], which penalizes late predictions exponentially harder than early ones (canonical penalty constants 13 and 10). Training-time loss functions that mirror this asymmetry have been proposed in recent work; the multiple-penalty-mechanism loss of [11] pairs RMSE with similarity-based late-prediction penalties and reports improved advanced-prediction probability on C-MAPSS subsets. We adopt asymmetric MSE with a $2\times$ late-prediction penalty as a methodological choice anchored on [14] and consistent with [11], rather than as an empirical contribution.

2.5 Where this work fits

Two observations motivate our contribution. First, recent attention and multi-scale-TCN architectures for RUL [7, 10, 18] are reported almost exclusively on the older C-MAPSS dataset, even though N-CMAPSS [1] has been available since 2021 and exposes substantially longer, multi-condition, variable-length flights. Second, the published N-CMAPSS literature itself [12, 6, 15, 5, 2] consists of a small number of competition entries and hybrid baselines, with no broad architecture sweep under matched preprocessing and training. Our work therefore benchmarks attention, multi-scale-TCN, transformer, WaveNet and RNN baselines on N-CMAPSS itself; the limited apples-to-apples N-CMAPSS attention literature is itself part of what makes the sweep informative. Where we cite C-MAPSS-only RMSE numbers (e.g. [7, 10, 18]), we do so as architectural precedent, not as a direct numerical baseline.

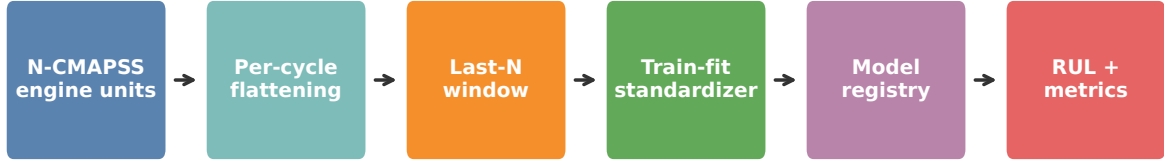
3 Methods

3.1 Dataset and task

We evaluate remaining-useful-life regression on N-CMAPSS [1], using the train, validation, and test splits exposed by the `rul_datasets` reader [13]. This is a simulation benchmark: N-CMAPSS combines a commercial-engine simulation model with real recorded flight profiles, and our experiments should not be read as validation on in-service engine failures. Each engine unit is represented as a list of per-cycle multivariate sensor sequences, and each cycle has one scalar RUL target. Unless otherwise noted, the reader caps RUL labels at 65 cycles and uses its default split, where the last 20% of the original training units are used for validation. For FD1 this yields 5 development units, 1 validation unit, and 4 test units. The default input has shape $(T, 32)$, where T is the retained temporal length and the 32 channels are 4 operating-condition variables, 14 physical sensors, and 14 virtual sensors. FD1 is the primary controlled benchmark and FD2 is used as a cross-subset robustness check.

3.2 Variable-length sequence preparation

Raw N-CMAPSS cycles can contain long, variable-length flight records. The training pipeline flattens the per-engine lists into per-cycle examples and, when a maximum sequence length is specified, keeps the last T timesteps of each cycle. Unless otherwise noted, the apples-to-apples benchmark



Common preprocessing, loss, callbacks, and evaluation harness across model families

Figure 1: Preprocessing and evaluation pipeline. Each engine is flattened into per-cycle examples, truncated to the last N timesteps, normalized with train-fit feature statistics, passed through the shared model registry, and evaluated with the same RUL metrics.

uses $T = 1000$. This last-window convention encodes the hypothesis that recent degradation behavior is more predictive of near-term RUL than distant early-flight history. After truncation, feature normalization is fit on the training split only and applied unchanged to validation and test examples, avoiding leakage from held-out engines.

3.3 Loss and metrics

All controlled benchmark rows in the main comparison use asymmetric MSE with late-prediction multiplier $\alpha = 2.0$:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \begin{cases} \alpha(\hat{y}_i - y_i)^2, & \hat{y}_i - y_i \geq 0, \\ (\hat{y}_i - y_i)^2, & \hat{y}_i - y_i < 0. \end{cases} \quad (1)$$

Here over-predicting RUL is treated as a late maintenance decision and is penalized more strongly than conservative under-prediction. This is a design choice motivated by the asymmetric PHM scoring function of Saxena et al. [14] and recent asymmetric-loss work [11]. Section 5 compares asymmetric MSE with symmetric MSE and asymmetric Huber under the same WaveNet FD1 setting.

Evaluation reports RMSE, R^2 , PHM score, and threshold accuracies. Accuracy@20 is the percentage of predictions within ± 20 cycles of the true RUL. The PHM score follows the standard asymmetric exponential form, with late errors penalized more sharply than early errors. We compare our own models only under matched preprocessing and training settings; externally published numbers are used as context for the N-CMAPSS result band, not as direct head-to-head baselines.

3.4 Model registry and benchmark harness

The repository currently registers 22 neural architectures under a common registry interface. The catalog spans a no-temporal MLP baseline, recurrent models [8, 4], dilated convolutional models [3, 16], CNN–RNN hybrids, attention models [17], transformer variants [7], and recent multi-scale TCN/fusion architectures [10, 18]. This shared interface lets the same preprocessing, optimizer, callbacks, loss, and metric code drive every architecture in the benchmark. Standard hyperparameters are units=64, dense units=32, dropout=0.2, Adam [9] with learning rate 0.001, 30 epochs for the controlled runs, and early stopping on validation loss.

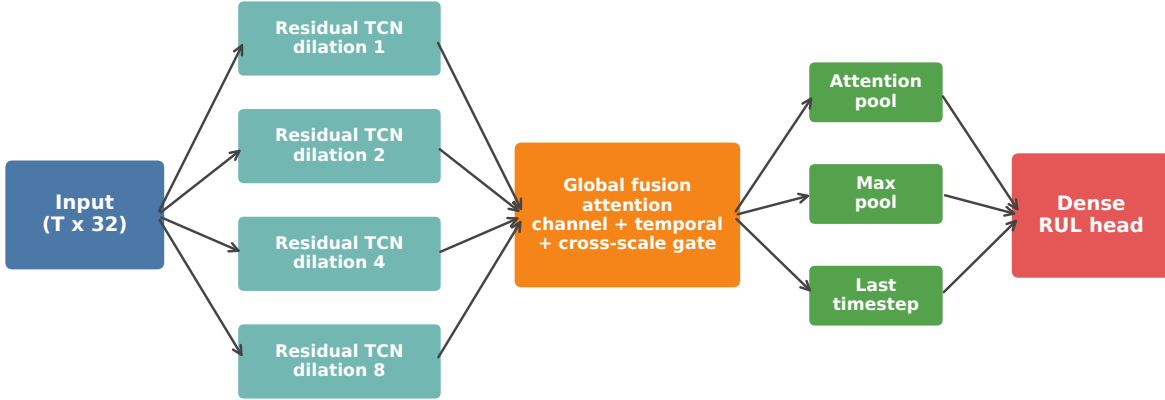


Figure 2: MSTCN architecture used in the benchmark. Parallel residual TCN branches cover four dilation scales before global fusion attention combines channel, temporal, and cross-scale information for the final RUL regression head.

3.5 MSTCN architecture

The flagship model is a multi-scale temporal convolutional network with global fusion attention (MSTCN), following the architectural family of attention-based multi-scale TCNs [18]. Given an input sequence $(T, 32)$, MSTCN applies four parallel residual TCN branches with dilation rates 1, 2, 4, and 8. Each branch captures a different temporal receptive field over the same sensor stream. The branch outputs are then passed through a global fusion attention block that combines channel attention, temporal attention, cross-scale concatenation, and a learned fusion gate. The fused representation is pooled over time and passed through a dense regression head to produce one scalar RUL estimate.

For the default configuration, MSTCN uses 64 filters per branch and a 32-unit dense head, yielding roughly 150k parameters. This is larger than some baselines, so our fixed-protocol claim is limited to “competitive” rather than “superior at matched parameter count.” An iso-parameter comparison has not yet been run.

4 Experiments

4.1 Experimental setup

The controlled benchmark uses N-CMAPSS FD1 as the primary subset and FD2 as a secondary cross-subset check. Unless otherwise noted, the apples-to-apples rows use the same data pipeline described in Section 3: last-window truncation, train-fit feature standardization, reader max RUL 65, Adam optimization [9], and the shared callback stack. The final controlled suite uses three seeds (42–44), with machine-readable metrics committed alongside the code. External comparison rows are cited only as contextual literature anchors, not as direct matched baselines.

Table 1: Architecture families in the benchmark harness.

Family	Registered models	Role in this paper
Baseline	MLP	No-temporal lower bound
RNN	LSTM [8], BiLSTM, GRU [4], BiGRU	Long-sequence recurrent baselines
Convolutional	TCN [3], WaveNet [16]	Dilated convolution baselines
Hybrid	CNN-LSTM, CNN-GRU, Inception-LSTM	Local feature extractor plus recurrence
Attention	Attention-LSTM, Transformer [17], ResNet-LSTM	Attention and transformer comparisons
Published fusion	MDFA, MDFA-paper, CNN-LSTM-Attention, CATA-TCN, TTSNet [10], ATCN, MSTCN [18]	Multi-scale / attention architectures
Transformer variants	STAR [7], sparse-transformer-BiGRU	Recent hierarchical/sparse attention

Table 2: Controlled FD1 results at $T=1000$ over seeds 42–44. All rows use the same preprocessing, reader max RUL 65, asymmetric MSE, and 30-epoch budget with early stopping. Std is over seeds.

Model	Seeds	RMSE mean	RMSE std	Mean R^2	Acc@20
WaveNet	3	6.868	0.299	0.8986	98.24
MSTCN	3	7.890	0.526	0.8659	97.75
CNN-GRU	3	17.168	8.801	0.2561	64.13

4.2 Architecture sweep

Table 1 summarizes the architecture families covered by the registry. The current codebase catalog contains 22 registered models, ranging from simple no-temporal baselines to multi-scale attention and transformer variants.

4.3 FD1 headline results

Table 2 reports the three-seed FD1 head-to-head at $T=1000$. WaveNet is the strongest and most stable model in this comparison ($\text{RMSE } 6.868 \pm 0.299$). MSTCN is stable and competitive ($\text{RMSE } 7.890 \pm 0.526$), but it does not beat WaveNet on FD1. CNN-GRU has one strong seed and two collapsed seeds, so its mean is much worse than its best run.

The FD1 result is therefore not a single-run leaderboard. WaveNet is robustly best among the three models tested here, MSTCN is a stable second cluster member, and CNN-GRU should not be described as a reliable top model despite its good seed-42 result.

4.4 FD2 robustness

On FD2, MSTCN is the best model by RMSE among the available apples-to-apples rows (T11: 6.495), with WaveNet close behind (T12: 6.739) and CNN-GRU degrading under the more variable flight envelope (T13: 19.812). WaveNet still has the highest Accuracy@20 on FD2 (T12: 99.50 versus T11: 99.01). Thus, the FD2 result supports the “top cluster” framing for MSTCN without establishing dominance across every metric.

4.5 Multiseed reproducibility

The three-seed suite was run specifically to bound run-to-run variance. Table 2 shows that the ranking changes when seed variance is included: WaveNet remains strong, MSTCN remains stable, and CNN-GRU is not reproducible under the same settings. This demonstrates why architecture rankings should not rely on a single seed.

Table 3: Canonical FD2 cross-subset results.

Row	Model	RMSE	R^2	PHM	Acc@20
T11	MSTCN	6.495	0.8897	0.5755	99.01
T12	WaveNet	6.739	0.8813	0.6756	99.50
T13	CNN-GRU	19.812	-0.0265	5.1463	59.41

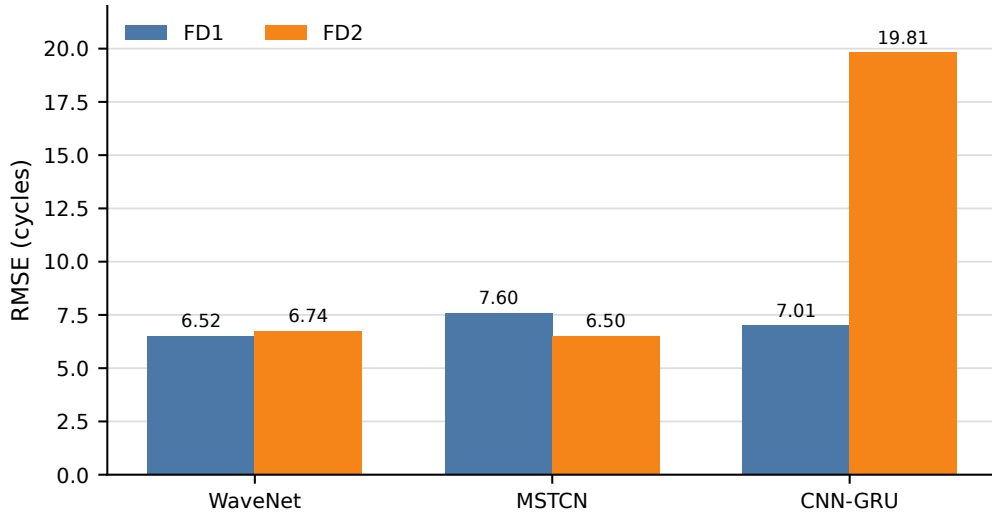


Figure 3: Controlled RMSE comparison on FD1 and FD2 for the three models available in both apples-to-apples runs. WaveNet is best on FD1, MSTCN is best by RMSE on FD2, and CNN-GRU degrades sharply on FD2.

4.6 Sequence-length evidence

Table 4 gives the clean sequence-length result requested by reviewers: one architecture (WaveNet), one training recipe, and three seeds. The sweep shows that short windows are effective across the tested range. The best mean is at $T=500$, with $T=250$ and $T=1000$ close enough that the result should be interpreted as a practical short-window regime rather than an exact optimum.

4.7 External comparison

The closest N-CMAPSS literature anchors are Solís-Martín et al. [15], who report validation RMSE 6.24 using a 161-cycle window, and Cohen et al. [5], who report RMSE 7.75 for ANN-Flux. Our three-seed FD1 WaveNet mean at $T=1000$ ($\text{RMSE } 6.868 \pm 0.299$), the best matched window-sweep mean at $T=500$ ($\text{RMSE } 6.690 \pm 0.173$), and the MSTCN FD1 mean ($\text{RMSE } 7.890 \pm 0.526$) are in the same broad N-CMAPSS numerical band. In addition, Table 5 reports the best single run found in a targeted seed sweep. WaveNet at $T=1000$ with seed 47 reaches RMSE 6.197 on FD1, slightly lower than the 6.24 N-CMAPSS anchor of Solís-Martín et al. Because the external rows use their own splits, inputs, and validation/test conventions, this is a repository benchmark result rather than a fully matched head-to-head claim. C-MAPSS-only results from STAR [7], TTSNet [10], and Xu et al. [18] are treated as architectural precedent rather than direct numerical baselines because the datasets and sequence regimes differ substantially.

We dissect these results across sequence length, architecture family, and loss form in Section 5.

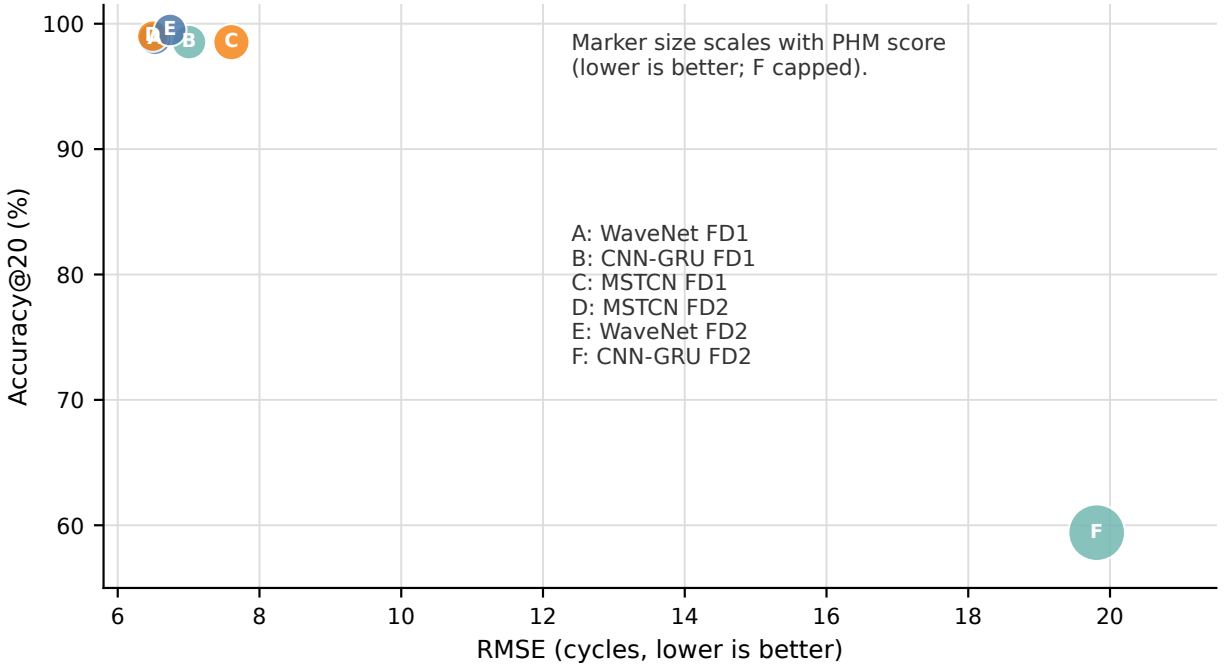


Figure 4: Accuracy–RMSE tradeoff for the controlled FD1/FD2 rows. Marker size scales with normalized PHM score, so lower and smaller points are better on the asymmetric PHM metric.

Table 4: Matched WaveNet sequence-length sweep on FD1 over seeds 42–44. All rows use asymmetric MSE, 32 features, reader max RUL 65, and the same training budget. Std is over seeds.

Sequence length	RMSE mean	RMSE std	Mean R^2	Acc@20
100	7.098	0.375	0.8916	97.95
250	6.880	0.531	0.8980	98.44
500	6.690	0.173	0.9038	98.24
1000	6.868	0.299	0.8986	98.24

Table 5: Best single FD1 run from the targeted seed sweep. This row is reported separately from the three-seed controlled tables because it is selected rather than averaged over seeds.

Model	Setting	Seed	RMSE	R^2	Acc@20
WaveNet	$T=1000$, asymmetric MSE, 32 features	47	6.197	0.9175	99.71

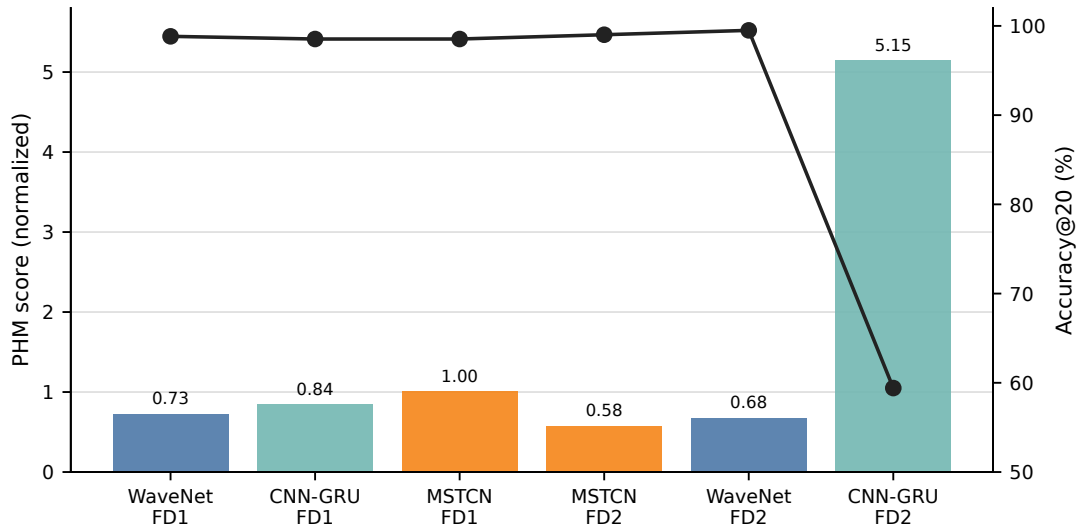


Figure 5: Normalized PHM score and Accuracy@20 for the controlled rows. Most top-cluster rows are tightly grouped on Accuracy@20, while PHM exposes additional separation because it penalizes late predictions asymmetrically.

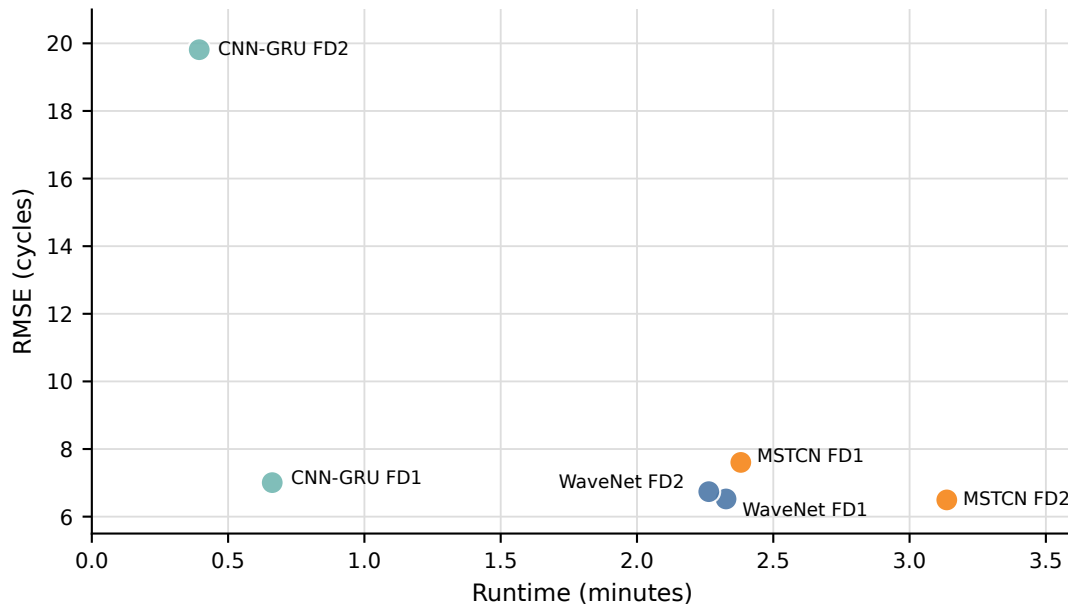


Figure 6: Runtime-RMSE tradeoff for the controlled FD1/FD2 rows. Runtime is measured by the benchmark script on the local machine and is reported as a practical engineering diagnostic, not as a hardware-normalized efficiency result.

5 Ablations and Discussion

This section interprets the benchmark against the three paper claims. Several experiments needed for causal attribution are absent, so we separate supported observations from missing ablations.

5.1 C1: sequence length

The matched WaveNet sweep supports the directional claim that short operational windows are effective on N-CMAPSS. Across seeds 42–44, the mean RMSE values are 7.098 at $T=100$, 6.880 at $T=250$, 6.690 at $T=500$, and 6.868 at $T=1000$. The differences among 250, 500, and 1000 timesteps are small relative to the scale of the task, while 100 timesteps is slightly worse. The supported conclusion is therefore a short-window regime, not a proof that exactly 1000 timesteps is optimal.

The remaining limitation is that this sweep stops at 1000 timesteps and does not include 2000, 5000, or full-flight inputs for the same architecture. Older full-flight baselines are much weaker, but those rows still conflate length with architecture and training details. We therefore report C1 as evidence for short operational windows, corroborated by short-window N-CMAPSS challenge submissions [12, 15], not as a complete full-flight ablation.

5.2 C2: multi-scale temporal attention

MSTCN is competitive but not uniformly dominant. In the final FD1 three-seed table, WaveNet has the best mean RMSE (6.868), MSTCN is second and stable (7.890), and CNN-GRU is unstable (17.168 mean RMSE because two seeds collapse). This contradicts any simple claim that MSTCN is the FD1 winner, but it supports treating MSTCN as a stable member of the stronger temporal-model group. On FD2, an older controlled run has MSTCN best by RMSE (T11: 6.495), ahead of WaveNet (T12: 6.739), but that FD2 observation has not yet been repeated with the same three-seed protocol.

The interpretation supported by the current results is that WaveNet is the robust FD1 winner among the tested models, MSTCN is a stable competitive model, and CNN-GRU should not be used as a reliable top baseline without additional tuning. The architectural attribution remains open. We have not yet ablated MSTCN’s multi-scale branches, removed its global fusion attention, or matched parameter counts against WaveNet and Transformer. Those experiments are required before claiming that the multi-scale or fusion-attention components themselves cause the observed performance.

5.3 C3: asymmetric loss

The project uses asymmetric MSE as its default loss, and the three-seed WaveNet comparison now gives preliminary empirical support for that choice. At $T=1000$, asymmetric MSE obtains mean RMSE 6.868 and mean Accuracy@20 98.24, symmetric MSE obtains mean RMSE 7.214 and mean Accuracy@20 97.65, and asymmetric Huber obtains mean RMSE 8.904 with high variance because one seed collapses. The loss is also aligned with the asymmetric PHM scoring tradition [14] and with recent cost-sensitive RUL loss design [11].

However, C3 is still preliminary. The comparison covers one architecture, one subset, one window length, and only three seeds. There is no α sweep over the late-penalty ratio and no architecture-wide loss comparison. We therefore frame asymmetric MSE as the best loss in this limited controlled suite, not as a general theorem about RUL loss design.

5.4 Open experiments

The next experiments are specific. Extending the matched WaveNet sweep to 2000, 5000, and full-flight inputs would test the C1 boundary. MSTCN variants with one dilation scale and with global fusion attention removed would test the C2 mechanism. Symmetric MSE, asymmetric MSE, and asymmetric Huber across multiple architectures, including an $\alpha \in \{1.0, 1.5, 2.0, 3.0\}$ sweep, would turn C3 from a limited FD1 observation into a broader empirical comparison.

6 Conclusion

We presented a deep-learning pipeline for variable-length N-CMAPSS turbofan RUL prediction and used it to compare recurrent, convolutional, hybrid, transformer, and multi-scale attention models under a common preprocessing, training, and evaluation harness. The controlled suite is a simulation study on FD1 with three seeds, reader max RUL 65, and explicit documentation of virtual-sensor use. WaveNet is the strongest controlled FD1 model tested here (RMSE 6.868 ± 0.299 at $T=1000$), MSTCN is stable and competitive (RMSE 7.890 ± 0.526), and CNN-GRU is not reliable because two seeds collapse. A matched WaveNet length sweep shows that short windows are effective across 100–1000 timesteps, with $T=500$ best by mean RMSE (6.690 ± 0.173). A separate targeted seed sweep gives the best individual run: WaveNet with $T=1000$, asymmetric MSE, 32 features, and seed 47 reaches RMSE 6.197, $R^2 = 0.9175$, and Accuracy@20 99.71 on FD1. Asymmetric MSE is modestly better than symmetric MSE in this limited suite, but broader loss and architecture ablations remain open. The code, configurations, and benchmark outputs provide a reproducible basis for extending this comparison; the selected best run should still be interpreted as a simulation benchmark rather than evidence for real-world commercial-engine deployment.

Limitations

- The sequence-length result is now matched for 100, 250, 500, and 1000 timesteps, but still needs 2000, 5000, and full-flight inputs under the same architecture.
- MSTCN component attribution is incomplete; the multi-scale branches and global fusion attention should be removed independently.
- The asymmetric-loss result is limited to WaveNet on FD1; it needs an α sensitivity sweep and replication across architectures and subsets.
- Evaluation is currently concentrated on FD1 and FD2; FD3–FD7 remain necessary for broad N-CMAPSS generalization claims.
- Comparisons to STAR, TTSNet, and Xu et al.’s MSTCN are architectural rather than apples-to-apples because those papers report primarily on C-MAPSS, not N-CMAPSS.

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